How to Execute Task 3: Mini Dialogue System (NLU → NLG inside an NLI Loop)

This experiment builds a simple chatbot that demonstrates:

- **NLU (Natural Language Understanding)**: Understands the user's intent.
- **NLG (Natural Language Generation)**: Generates an appropriate response.
- **NLI Loop (Natural Language Interaction)**: Continuously interacts with the user until they exit.

Step 1: Install Python

Check Python installation:

```
python --version
```

or

```
python3 --version
```

If Python is not installed, download it from:

https://www.python.org/downloads/

Step 2: Create the Program

Open a text editor and save the code as:

```
mini_dialogue_system.py
```

Paste the following code:

```
import re

def understand(text):
    # NLU
    text = text.lower()

    intent = "book_flight" \
        if re.search(r"\b(book|fly|flight)\b", text) \
        else "unknown"

     slots = re.findall(
         r"\b(from|to)\s+(\w+)",
         text
     )
```

```
15
16      return intent, slots
17
18
19  def generate(intent, slots):
20      # NLG
21
22      if intent == "book_flight":
23          if slots:
24              return (
25                  f"Searching flights for "
26                  f"{slots[0][1].title()} ..."
27              )
28
29          return (
30              "Sure. Where are you flying from, "
31              "and to?"
32          )
33
34      return (
35          "Sorry, I can only help with flights."
36      )
37
38
39  # NLI Loop
40
41  while True:
42      user = input("You: ")
43
44      if user.lower() in ["bye", "exit"]:
45          break
46
47      intent, slots = understand(user)
48
49      print(
50          "Bot:",
51          generate(intent, slots)
52      )
```

Step 3: Run the Program

Open Terminal / Command Prompt:

```
python mini_dialogue_system.py
```

Step 4: Interact with the Bot

Example 1

```
You: I want to book a flight
Bot: Sure. Where are you flying from, and to?
```

Example 2

```
You: book flight from Mumbai
Bot: Searching flights for Mumbai...
```

Example 3

```
You: I need a hotel
Bot: Sorry, I can only help with flights.
```

Exit

```
You: bye
```

Program terminates.

Understanding the Code

1. NLU Phase

Function:

```
understand()
```

Purpose:

- Converts input to lowercase.
- Detects intent.
- Extracts locations.

Example:

Input:

```
book flight from Delhi to Chennai
```

Output:

```
intent = "book_flight"

slots = [
    ('from', 'Delhi'),
    ('to', 'Chennai')
]
```

2. Intent Recognition

```
re.search(
    r"\b(book|fly|flight)\b",
    text
)
```

Recognizes keywords:

```
book
fly
flight
```

If found:

```
intent = "book_flight"
```

Otherwise:

```
intent = "unknown"
```

3. Slot Filling

```
re.findall(
    r"\b(from|to)\s+(\w+)",
    text
)
```

Extracts travel information.

Example:

```
book flight from Delhi to Chennai
```

Returns:

```
[
 ('from', 'Delhi'),
 ('to', 'Chennai')
```

```
]
```

These are called **slots**.

---

4. NLG Phase

Function:

```
generate()
```

Generates a response based on:

- Intent
- Extracted slots

Example:

```
generate(
 "book_flight",
 [('from','Delhi')]
)
```

returns:

```
Searching flights for Delhi...
```

---

5. NLI Loop

```
while True:
```

Keeps the conversation running.

```
You → NLU → Intent/Slots
           ↓
         NLG
           ↓
          Bot
```

until:

```
bye
```

or

```
exit
```

is entered.

Sample Transcript

```
You: I want to book a flight
Bot: Sure. Where are you flying from, and to?

You: from Mumbai to Delhi
Bot: Searching flights for Mumbai ...

You: book a hotel
Bot: Sorry, I can only help with flights.

 You: bye
```

Viva Questions

1. What is NLU?

Natural Language Understanding is the process of extracting meaning from user input.

2. What is NLG?

Natural Language Generation creates human-readable responses.

3. What is Intent Recognition?

Identifying the user's goal.

Example:

```
Book a flight
```

→ book_flight

4. What are Slots?

Important pieces of information extracted from text.

Example:

```
from Delhi
to Chennai
```

5. What is the NLI Loop?

The continuous cycle:

```
User Input
     ↓
NLU
```

```
      ↓
Intent & Slots
      ↓
NLG
      ↓
Bot Response
       ↓
 User Input
```

Conclusion

This experiment demonstrates a basic conversational AI system using:

- **NLU** for intent detection and slot extraction
- **NLG** for response generation
- **NLI Loop** for continuous interaction

It illustrates the core architecture used in chatbots, virtual assistants, customer-support bots, and flight-booking assistants in Natural Language Processing (NLP).